Applied Statistics

Lecture 8:
Bootstrap Confidence Intervals

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21st October 2025

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Learning Objectives for this lecture

- Recognise different approaches for computing confidence intervals
- Explain the differences between Neyman's construction, percentile method, as well as BC and BCa methods.
- Apply and implement confidential intervals in applications
- Relate the Bayes credibility intervals, and confidence density to confidence intervals

Outline of lecture

Introduction

Neyman's construction

The percentile method

Bias-corrected confidence intervals

Second-order-corrected confidence intervals

Objective Bayes intervals

Introduction

- The familiar "**standard confidence intervals**"

$$\hat{\theta} \pm \widehat{\text{se}} \tag{1}$$

- The standard intervals are symmetric around $\hat{\theta}$ and not very accurate
- **The goal:** automatic estimation of asymmetric confidence intervals accurately

Recap: Measurement of height

Assume that we have 10 repeated measurements of a persons height:

```
1.983257, 1.974929, 2.001295, 1.983005, 2.021214,
2.060859, 1.975049, 1.975376, 1.984431, 1.944212
```

- The mean $\bar{x}_{10} = 1.990363$ is a **point estimate**
- It is **close to** but not (necessarily) equal to the true height θ.
- **Confidence intervals** quantify what we mean by **close to**.

We want to make statements like: we are 95% **confident** that $\theta \in (1.967733, 2.012993)$.

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Exercise

Why can't I say: $\theta \in (1.967733, 2.012993)$ with probability 95%?

Recap: Confidence intervals

Definition (Confidence intervals, Dekking et al. p. 343)

Suppose a dataset $x_1 \ldots, x_n$ is given, modelled as realization of random variables $X_1, \ldots, X_n$. Let θ be the parameter of interest, and γ a number between 0 and 1. If there exist sample statistics $L_n = g(X_1, \ldots, X_n)$ and $U_n = h(X_1, \ldots, X_n)$ such that

$$P(L_n < \theta < U_n) = \gamma$$

for every value of θ, then

$$(l_n, u_n),$$

where $l_n = g(x_1, \ldots, x_n)$ and $u_n = h(x_1 \ldots, x_n)$, is called a $100\gamma\%$ **confidence interval for** θ. The number γ is called the confidence level.

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Neyman's construction

Example

Remember the earlier example with student score data where the correlation coefficent between the "mechanics" and "vectors" was estimated, where

$$\hat{\theta} = 0.498. \tag{2}$$

Question: How to evaluate the confidence intervals of the estimate.

Table 3.1 *Scores from two tests taken by 22 students,* **mechanics** *and* **vectors**.

	1	2	3	4	5	6	7	8	9	10	11
mechanics	7	44	49	59	34	46	0	32	49	52	44
vectors	51	69	41	70	42	40	40	45	57	64	61

	12	13	14	15	16	17	18	19	20	21	22
mechanics	36	42	5	22	18	41	48	31	42	46	63
vectors	59	60	30	58	51	63	38	42	69	49	63

Neyman's construction (2)

Example

Solution:

- Assuming the bivariate normal model, there is an analytic form for the normal correlation coefficient density $f_{\hat{\theta}}(r)$.
- The CIs are defined as the numerical solutions to

$$\int_{\hat{\theta}}^{1} f_{\hat{\theta}_{\text{lo}}}(r)dr = \alpha \qquad (3)$$

and

$$\int_{-1}^{\hat{\theta}} f_{\hat{\theta}_{\text{up}}}(r)dr = \alpha \qquad (4)$$

where $\alpha = 0.025$.

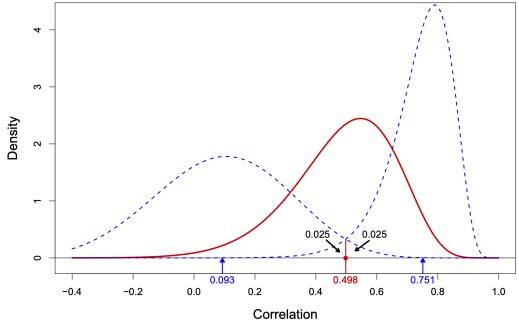


Figure 11.1 The solid curve is the normal correlation coefficient density $f_{\hat{\theta}}(r)$ (3.11) for $\hat{\theta} = 0.498$, the MLE estimate for the student score data; $\hat{\theta}(\text{lo}) = 0.093$ and $\hat{\theta}(\text{up}) = 0.751$ are the endpoints of the 95% confidence interval for θ, with corresponding densities shown by dashed curves. These yield tail areas 0.025 at $\hat{\theta}$ (11.6).

Transformation invariance

- In general, we want that the confidence interval are **transformation invariant**
- This means that for a monotonically increasing function m for parameters

$$\phi = m(\theta) \tag{5}$$

  the level-α confidence intervals are equivalent i.e., the levels map via the function m.

- Neyman's construction is transformation invariant in this sense

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Recap: Empirical bootstrap for the studentized mean

What do we do if we have a data set $x_1, x_2, \ldots, x_n$ that is a realization of a random sample from some distribution F that is either not normal or unknown?

Definition (Dekking et al. p. 351)

Given a dataset $x_1, x_2, \ldots, x_n$, determine its empirical distribution function F_n as an estimate of F. The expectation corresponding to F_n is $\mu^* = \bar{x}_n$.

1. Generate a bootstrap dataset $x_1^*, x_2^*, \ldots, x_n^*$ from F_n.
2. Compute the studentized mean for the bootstrap dataset

$$t^* = \frac{\bar{x}_n^* - \bar{x}_n}{s_n^*/\sqrt{n}}$$

where $\bar{x}_n^*$ and s_n^* are the sample mean and sample standard deviation of $x_1^*, x_2^*, \ldots x_n^*$.

Repeat steps 1 and 2 many times.

Recap: Empirical bootstrap for the studentized mean

Definition (Dekking et al. p. 351)

Given a dataset $x_1, x_2, \ldots, x_n$, determine its empirical distribution function F_n as an estimate of F. The expectation corresponding to F_n is $\mu^* = \bar{x}_n$.

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Repeat steps 1 and 2 many times.

From bootstrap datasets we can determine c_l^* and c_u^* such that

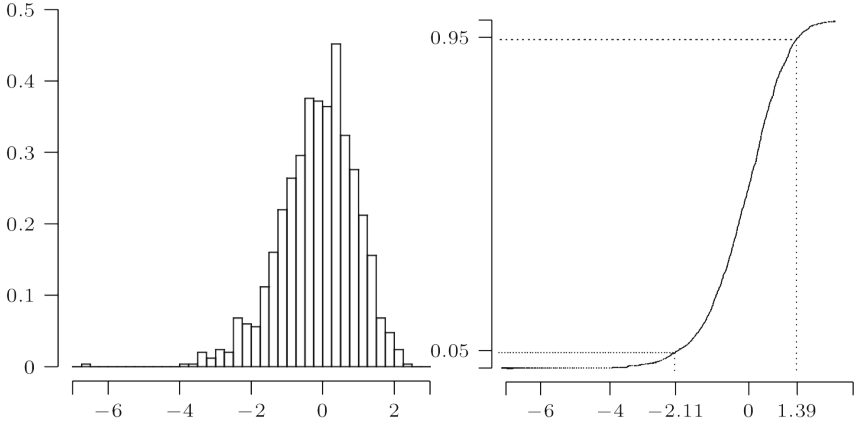
$$P\left(c_l^* < \frac{\bar{x}_n^* - \bar{x}_n}{s_n^*/\sqrt{n}} < c_u^*\right) \approx 1 - \alpha$$

which gives the $100(1-\alpha)\%$ confidence interval for μ

$$\left(\bar{x}_n - c_u^* \frac{s_n}{\sqrt{n}} \;\;,\;\; \bar{x}_n - c_l^* \frac{s_n}{\sqrt{n}}\right).$$

Recap: Empirical bootstrap simulation for the studentized mean

This means that c_l^* and c_u^* is determined from the sampling distribution of t^*



The percentile method

- Empirical bootstrap for the studentised mean is an example of the **percentile method**
  The idea in summary
  1. Generate bootstrap samples of a selected statistic
  2. Define the *bootstrap cumulative distribution function*

$$\hat{G}(t) = \#\{\hat{\theta}^{*b} \leq t\}/B \tag{6}$$

  3. Solve the equation

$$\hat{\theta}^{*(\alpha)} = G^{-1}(\alpha) \tag{7}$$

  for the αth percentile points.

- The percentile intervals are **transfromation invariant**.

The percentile method (2)

Properties

- Does not require knowing the transformation to normality $\hat{\phi} = m(\hat{\theta})$, it only assumes its existence.
- If the transformation exists, correct intervals in the Fisherian sense
- Standard intervals work fine once in the right frame—but they are not transformation invariant.
- Bootstrap sample sizes $B \approx 2000$ required
- There are even better methods than percentile methods: **BC** and **BCa**

BC and BCa methods

- Ideally, the percentile method is applied for the normal bootstrapped statistic

$$\hat{\phi}^* \sim N(\hat{\phi}, \sigma^2) \tag{8}$$

  which is an **unbiassed estimator** with **constant variance** generated by the transformation $\hat{\phi} = m(\hat{\theta})$.

- If we do not have a direct access to unbiassed, constantly varying bootstrapped statistics, we may however **remove the bias** and **take into account the changing variance**.

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Bias-corrected (BC) confidence intervals

Definition

Let $\hat{\theta}^{*1}, \hat{\theta}^{*2}, \ldots, \hat{\theta}^{*B}$ represent parametric or non-parametic bootstrap replications of the statistic $\hat{\theta}$, and p_0 be the proportion of replications less than $\hat{\theta}$, or,

$$p_0 = \frac{\#\left\{\hat{\theta}^{*b} \leq \hat{\theta}\right\}}{B}. \tag{9}$$

Let the bias correction value

$$z_0 = \Phi^{-1}(p_0), \tag{10}$$

where Φ^{-1} is the inverse of the standard normal cdf. The bias corrected (BC) level-α confidence interval endpoint is then

$$\hat{\theta}_{BC}(\alpha) = \hat{G}^{-1}\left[\Phi(2z_0 + z^{(\alpha)})\right], \tag{11}$$

where $\hat{G}$ is the bootstrap cdf and $z^{(\alpha)} = \Phi^{-1}(\alpha)$.

BC properties

- More generally, if there is a monotonic transformation $\phi = m(\theta)$ such that for any θ

$$\hat{\phi} \sim N(\phi - z_0\sigma, \sigma^2), \tag{12}$$

  where σ, z_0 constants, then BC endpoints can be shown accurate.

- Correct probabilities in the Fisherian sense.

- Note that there is no additional computational burden.

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Second-Order accuracy

- Standard confidence intervals, percentage method, and BC method have the actual coverage

$$\Pr_\theta \left\{ \theta \leq \hat{\theta}_{\text{standard}}[\alpha] \right\} = \alpha + \frac{c_1}{\sqrt{n}}, \tag{13}$$

  where c_1 is a problem dependent constant.

- Second-order accurate methods

$$\Pr_\theta \left\{ \theta \leq \hat{\theta}_{\text{2nd}}[\alpha] \right\} = \alpha + \frac{c_2}{n}, \tag{14}$$

  where c_2 is also a constant.

- BC method does not yield 2nd order accuracy due to the assumption of **constant standard error** in the model $\hat{\phi} \sim N(\phi - z_0\sigma, \sigma^2)$, $\hat{\phi} = m(\hat{\theta})$.

- Updating the assumption to monotonic transformation

$$\hat{\phi} \sim N(\phi - z_0\sigma_\phi, \sigma_\phi^2), \quad \sigma_\phi = 1 + a\phi, \tag{15}$$

  where a is the **acceleration** constant (must be computed separately).

Bias-corrected and accelerated (BCa) confidence intervals

Definition

Let $\hat{\theta}^{*1}, \hat{\theta}^{*2}, \ldots, \hat{\theta}^{*B}$ represent parametric or non-parametic bootstrap replications of the statistic $\hat{\theta}$, and p_0 be the proportion of replications less than $\hat{\theta}$. Let the bias correction value

$$z_0 = \Phi^{-1}(p_0), \tag{16}$$

where Φ^{-1} is the inverse of the standard normal cdf. The bias corrected and accelerated (BCa) level-α confidence interval endpoint is then

$$\hat{\theta}_{BC}(\alpha) = \hat{G}^{-1}\left[\Phi\left(z_0 + \frac{z_0 + z^{(\alpha)}}{1 - a(z_0 + z^{(\alpha)})}\right)\right], \tag{17}$$

where $\hat{G}$ is the bootstrap cdf and $z^{(\alpha)} = \Phi^{-1}(\alpha)$.

It can be shown that under the existence of monotonic transformation m and the constant a, then the BCa are correct in the Fisherian sense.

Example: Student Score Correlation

Example

Table 11.1 *Bootstrap confidence limits for student score correlation,* $\hat{\theta} = 0.498$, $n = 22$. *Parametric exact limits from Neyman's construction as in Figure 11.1. The BC and BCa methods are discussed in the next two sections;* (z_0, a), *two constants required for BCa, are* $(-0.055, 0.005)$ *parametric, and* $(0.000, 0.006)$ *nonparametric.*

	Parametric		Nonparametric	
	.025	.975	.025	.975
1. Standard	.17	.83	.18	.82
2. Percentile	.11	.77	.13	.76
3. BC	.08	.75	.13	.76
4. BCa	.08	.75	.12	.76
Exact	**.09**	**.75**		

Example: Poisson parameter confidence intervals

Example

Table 11.2 *Nominal 95% central confidence intervals for Poisson parameter θ having observed $\hat{\theta} = 10$; actual tail areas above and below $\hat{\theta} = 10$ defined as in Figure 11.1 (atom of probability split at 10). For instance, lower standard limit 3.80 actually puts probability 0.004 above 10, rather than nominal value 0.025. Bias correction value z_0 (11.32) and acceleration a (11.38) both equal 0.050.*

	Nominal limits		Tail areas	
	.025	.975	Above	Below
1. Standard	3.80	16.20	.004	.055
2. %ile	4.18	16.73	.007	.042
3. BC	4.41	17.10	.010	.036
4. BCa	5.02	17.96	.023	.023
Exact	**5.08**	**17.82**	**.025**	**.025**

Remarks

- The percentile, BC, and BCa methods for cumulative correction of non-normality, bias, and acceleration, respectively.
- The percentile and BC methods are completely automatic, BCa has the additional parmaeter a that needs to be estimated (by e.g. Jackknife).
- CI methods require ca 2000 bootstrap replications, about ten times as many as with SE.
- So far we have considered univariate problems—there are particular callenges in higher dimensions (e.g. to handling biasses).

Quiz

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Objective Bayes intervals

Objective Bayes intervals

Definition

Let $f_\theta(\hat{\theta})$ be a one parameter family of densities and $g(\theta)$ be a prior density. The Bayes α **credible interval** $C(\theta|\hat{\theta}) = [a(\theta), b(\theta)]$ spans the central α region of the posterior $g(\theta|\hat{\theta}) = f_\theta(\hat{\theta})g(\theta)/\int f_\theta(\hat{\theta})g(\theta)d\theta$ so that

$$\int_{a(\hat{\theta})}^{b(\hat{\theta})} g(\theta|\hat{\theta})d\theta = \alpha, \tag{18}$$

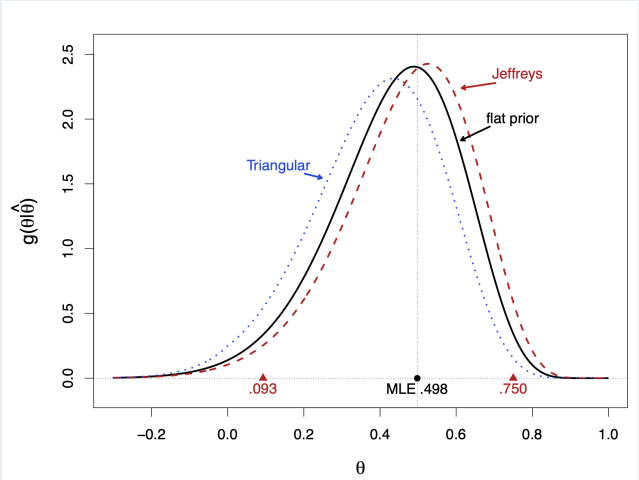
where the posterior probability is divided evenly to each tail region.

- We are particularly interested in **matching priors** so that the credible intervals approximate Neyman confidence intervals
- Example: Jeffrey's prior

Student Score Correlation Revisited

Example

Neumann intervals are near exact match with Bayes credible intervals with Jeffreys prior!



Confidence density

Posterior densities with matching prior have a frequentist correspondent.

Definition

Let $\hat{\theta}_x[\alpha]$ denote the upper confidence limit so that $\Pr\left\{\theta \leq \hat{\theta}_x[\alpha]\right\} = \alpha$. Denoting $\alpha = \alpha(\theta)$, and assuming that the function $\alpha(\theta)$ is differentiable on the domain of θ, the confidence density of θ is

$$\tilde{g}_x(\theta) = \frac{d\alpha}{d\theta}. \tag{19}$$

Note that this is not controverial as long as we remember that this is not a random quantity in the frequentist sense but rather an interval density that varies in the funtion of x.

Confidence density (2)

Example

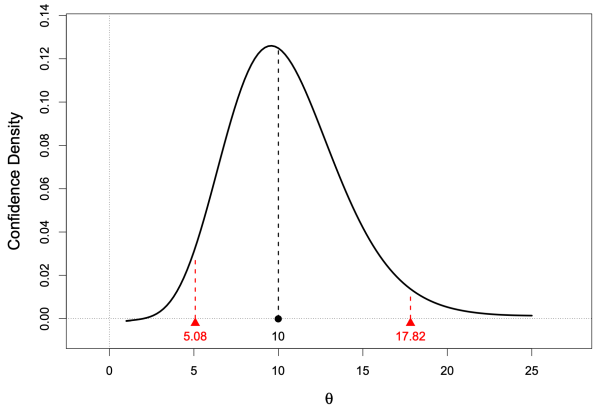


Figure 11.6 Confidence density (11.62) for Poisson parameter θ having observed $\hat{\theta} = 10$. There is area 0.95 under the curve between 5.08 and 17.82, as in Table 11.2, and areas 0.025 in each tail.

Confidence density (3)

- If the prior gives a perfect match to the confidence interval system, the condifence density are equivalent to the **posterior density** with the matching prior!
- Also for the **percentile method** the bootstrap density is the confidence density.
- For BCa, the confidence density can be obtained by reweighting.

Confidence density (4)

Example

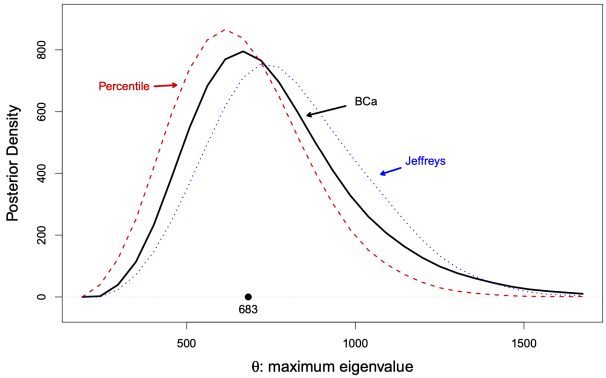


Figure 11.7 Confidence densities for the maximum eigenvalue parameter (11.71), using a multivariate normal model (11.70) for the student score data. The dashed red curve is the percentile method, solid black the BCa (with $(z_0, a) = (0.178, 0.093)$). The dotted blue curve is the Bayes posterior density for θ, using Jeffreys' prior (11.72).